



Practice Programs

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Program 1: Write a C program to print the following text message each time it is run.
'To C, or not to C: that is the question.'

pun.c :

```
1  #include <stdio.h>
2
3  int main(void) {
4      printf("To C, or not to C: that is the question.\n");
5      return 0;
6  }
```

Program 2: Write a C program to demonstrate simple arithmetic.

simpleArithmetic.c

```
1  #include<stdio.h>
2
3  int main(){
4      int a, b, c;
5
6      printf("Please enter two numbers : ");
7      scanf("%d %d", &a, &b); //ok
8
9      c = a + b; //addition
10     printf("The Sum is %d\n", c);
11
12     c = a - b; //subtraction
13     printf("The Difference is %d\n", c);
14
15     c = a * b; //multiplication
16     printf("The Product is %d\n", c);
17
18     c = a / b;
19     printf("The Division is %d\n", c);
20     return 0;
21 }
```

Program 3: Write a C program to calculate Simple Interest.

simpleInterestCalculator.c

```

1  #include<stdio.h>
2
3  int main(){
4      int numberOfYears;
5      float principleAmount, rateOfInterest, simpleInterest;
6
7      printf("Enter Principle Amount\n", principleAmount);
8      scanf("%f", &principleAmount);
9      printf("Enter number of year\n", numberOfYears);
10     scanf("%d", &numberOfYears);
11     printf("Enter Rate of Interest\n", rateOfInterest);
12     scanf("%f", &rateOfInterest);
13
14     simpleInterest = (principleAmount * numberOfYears * rateOfInterest) / 100;
15     printf("Simple interest is %f\n", simpleInterest);
16
17     return 0;
18 }
19

```

Program 4: Write a C program to calculate Simple EMI.

simpleEMICalculator.c

```

1  #include <stdio.h>
2  #include <math.h>
3
4  int main() {
5      float principle, rate, time, emi;
6
7      printf("Enter principle amount: ");
8      scanf("%f", &principal);
9
10     printf("\nEnter rate of interest: ");
11     scanf("%f", &rate);
12
13     printf("\nEnter time in years: ");
14     scanf("%f", &time);
15
16     rate = rate / (12 * 100); /*one month interest*/
17     time = time * 12; /*one month period*/
18
19     emi= ( principal * rate * pow(1 + rate,time)) / (pow(1 + rate,time) - 1);
20
21     printf("Monthly EMI is= %f\n", emi);
22
23     return 0;
24 }
25

```

Program 5: Write a C program to calculate Student Percentage of 5 subjects

studentPercentageCalculator.c

```

1  #include<stdio.h>
2
3  int main() {
4      int a,b,c,d,e;
5      float total, average, percentage;
6
7      printf("Enter marks of 5 subjects : \n");
8      scanf("%d %d %d %d %d", &a, &b, &c, &d, &e);
9
10     total = a + b + c + d + e;
11     average = total / 5;
12     percentage = (total / 500) * 100;
13
14     printf("Total marks = %f\n", total);
15     printf("Average marks = %f\n", average );
16     printf("Net percentage = %f\n", percentage);
17     return 0;
18 }
19

```

Program 6: Write a program taking user input of basic salary and calculating gross salary that includes basic salary, 50% DA and 40% HRA.

grossSalaryCalculator.c

```

1  #include<stdio.h>
2
3  void main() {
4      float basic,hra,da,gross;
5
6      printf("Enter the Basic Salary : $");
7      scanf("%f", &basic);
8
9      hra = 40 * basic / 100;
10     da = 50 * basic / 100;
11     gross = basic + hra + da;
12
13     printf("Gross Salary is $%f", gross);
14 }
15

```

Program 7: Write a program to calculate area of Square, Triangle & Rectangle using user inputs of shape data.

areaOfShapes.c

```
1  #include <stdio.h>
2
3  void main() {
4      float area;
5      float side; //For square
6      float base, height; //For triangle
7      float length, breadth; //For rectangle
8
9      //Area of square
10     printf("Enter the side of square in cms: ");
11     scanf("%f", &side);
12     area = side * side;
13     printf("Area of square with sides %.2f cms is %f\n", side, area);
14
15     //Area of Triangle
16     printf("Enter the base of triangle in cms : ");
17     scanf("%f", &base);
18     printf("Enter the height of triangle in cms : ");
19     scanf("%f", &height);
20     area = 0.5 * base * height;
21     printf("Area is of triangle with base %.2fcms and height %.2fcms is %fcms\n", base, height, area);
22
23     //Area of Rectangle
24     printf("Enter the length of rectangle in cms : ");
25     scanf("%f", &length);
26     printf("Enter the breadth of rectangle in cms : ");
27     scanf("%f", &breadth);
28     area = length * breadth;
29     printf("The area of rectangle with length %.2fcms and breadth %.2fcms is %fcms\n", length, breadth, area);
30 }
```


Program 8: Write a program to calculate Area & Circumference of a Circle, take user inputs.

areaAndCircumference.c

```

1  #include<stdio.h>
2
3  int main() {
4      float radius, area, circumference;
5
6      printf("Enter the radius of Circle in cms : ");
7      scanf("%f", &radius);
8
9      area = 3.14 * radius * radius; //Area of circle
10     circumference = 2 * 3.14 * radius; //Circumference of circle
11
12     printf("Area of circle is %f\n", area);
13     printf("Circumference of circle is %f\n", circumference);
14     return 0;
15 }
16

```

Program 9: Write a program to calculate Dimensional weight of a box to help shipping companies charge accordingly.

dimensionalWeight.c

```

1  /*The problem is about calculating the dimensional weight of a box,
2   which shipping companies use to charge based on space taken rather
3   than actual weight. The formula divides the box's volume by 166,
4   but since integer division in C truncates decimals (rounds down),
5   we adjust by adding 165 before dividing to properly round up.*/
6
7  #include <stdio.h>
8
9  int main(void) {
10     int height, length, width, volume, weight;
11
12     printf("Enter height of the box: ");
13     scanf("%d", &height);
14     printf("Enter length of the box: ");
15     scanf("%d", &length);
16     printf("Enter width of the box: ");
17     scanf("%d", &width);
18
19     volume = height * length * width;
20     weight = (volume + 165) / 166;
21
22     printf("Volume of box is %d (cubic inches)\n", volume);
23     printf("Dimensional weight of the box is %d pounds\n", weight);
24 }

```


Program 10: Write a program to convert the user given temperature in Celsius(*C) into Fahrenheit(*F).

celsiusToFahrenheit.c

```

1  /* Converts a Celsius temperature to Fahrenheit */
2
3  #include<stdio.h>
4
5  #define FREEZING_PT 32.0f
6  #define SCALE_FACTOR (9.0f / 5.0f)
7
8  int main() {
9      float celsius, fahrenheit;
10
11     printf("Enter the Temperature in Celcius : ");
12     scanf("%f", &celsius);
13
14     fahrenheit = SCALE_FACTOR * celsius + FREEZING_PT;
15
16     printf("Temperature in Fahernheit is %.1f F", fahrenheit);
17
18     return 0;
19 }
```

Program 11: Write a program to convert the user given temperature in Fahrenheit(*F) into Celsius(*C).

fahrenheitToCelsius.c

```

1  /* Converts a Fahrenheit temperature to Celsius */
2
3  #include <stdio.h>
4
5  #define FREEZING_PT 32.0f
6  #define SCALE_FACTOR (5.0f / 9.0f)
7
8  int main() {
9      float fahrenheit, celsius;
10
11     printf("Enter Fahrenheit temperature: ");
12     scanf("%f", &fahrenheit);
13
14     celsius = (fahrenheit - FREEZING_PT) * SCALE_FACTOR;
15
16     printf("Celsius equivalent: %.1f\n", celsius);
17
18     return 0;
19 }
```

Program 12: Write a program to convert the user given temperature in Celsius(*C) to Kelvin(*K).

celsiusToKelvin.c

```

1  /* Converts a Celsius temperature to Kelvin */
2
3  #include <stdio.h>
4
5  #define SCALE_FACTOR 273.15
6
7  int main()
8  {
9      float celsius, kelvin;
10
11     printf("Enter the Temperature in Celcius : ");
12     scanf("%f", &celsius);
13
14     kelvin = celsius + SCALE_FACTOR;
15
16     printf("Temperature in Kelvin is %.2f k\n", kelvin);
17     return 0;
18 }
```

Program 13: Write a program to convert the user given temperature in Kelvin(*K) into Celsius(*C).

kelvinToCelsius.c

```

1  /* Converts a Kelvin temperature to Celsius */
2
3  #include <stdio.h>
4
5  #define SCALE_FACTOR 273.15
6
7  int main()
8  {
9      float celsius, kelvin;
10
11     printf("Enter the Temperature in Kelvin : ");
12     scanf("%f", &kelvin);
13
14     celsius = kelvin - SCALE_FACTOR;
15
16     printf("Temperature in Celsius is %.2f C\n", celsius);
17     return 0;
18 }
```

Program 14: Write a program to accept the distance between two cities in kilometres from the user. Calculate and display this distance in meters, feet, centimetres and inches

distanceConverter.c

```

1  #include<stdio.h>
2  #include<conio.h>
3
4  void main() {
5      float km, mt, inch, ft, cm;
6
7      printf("Enter the distance between two cities in kilometers : ");
8      scanf("%f", &km);
9
10     mt = km * 1000;
11     ft = mt * 3.33;
12     cm = mt * 100;
13     inch = ft * 12;
14
15     printf("The distance in meters is = %.2f mts.\n", mt);
16     printf("The distance in feets is = %.2f ft.\n", ft);
17     printf("The distance in centimeters is = %.2f cms.\n", cm);
18     printf("The distance in inchs is = %.2f inches.\n", inch);
19 }
```

Program 15: Write a program to demonstrate bitwise operator on two integers.

bitwiseCalculation.c

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int a = 14, b = 7;
6
7      printf("Bitwise NOT of a (~a): %d\n", ~a);
8      printf("Bitwise NOT of b (~b): %d\n\n", ~b);
9
10     printf("a & b = %d\n", a & b);
11
12     printf("a | b = %d\n", a | b);
13
14     printf("a ^ b = %d\n\n", a ^ b);
15
16     printf("a << 1 = %d\n", a << 1);
17     printf("b << 1 = %d\n\n", b << 1);
18
19     printf("a >> 1 = %d\n", a >> 1);
20     printf("b >> 1 = %d\n", b >> 1);
21
22     return 0;
23 }
```

Program 16: Write a program to demonstrate increment operator on an integer.

incrementOperator.c

```

1 // Increment operator
2
3 #include<stdio.h>
4 int main()
5 {
6     int a= 5;
7     int b, c, d;
8
9     printf("Value of a = %d\n", a);
10
11     b = ++a;
12     c = a++;
13     d = ++a;
14
15     printf("Value of b using ++a = %d\n", b);
16     printf("Value of c using a++ = %d\n", c);
17     printf("Value of d using ++a = %d\n", d);
18
19     return 0;
20 }
```

Program 17: Write a program to demonstrate decrement operator on an integer.

decrementOperator.c

```

1 // Decrement operator
2
3 #include<stdio.h>
4
5 int main() {
6     int a= 8;
7     int b, c, d;
8
9     printf("Value of a = %d\n", a);
10
11     b = --a;
12     c = a--;
13     d = --a;
14
15     printf("Value of b using --a = %d\n", b);
16     printf("Value of c using a-- = %d\n", c);
17     printf("Value of d using --a = %d\n", d);
18
19     return 0;
20 }
```

Program 18: Write a program to calculate size of variables.

calculateSizeOfOperator.c

```

1  /* sizeof() operator in c is used to calculate the size of a variable
2  used inside the program and return the size in integer in the form
3  of memory bytes*/
4
5  #include<stdio.h>
6
7  int main()
8  {
9      int a = 6;
10     float b = 8.765f;
11     long long int c = 15050603LL;
12     double d = 78945.321654;
13     char ch = 'X';    // Capital X
14
15     printf("Size of a : %d Bytes\n", sizeof(a));
16     printf("Size of b : %d Bytes\n", sizeof(b));
17     printf("Size of c : %d Bytes\n", sizeof(c));
18     printf("Size of d : %d Bytes\n", sizeof(d));
19     printf("Size of X : %d Bytes\n", sizeof(ch));
20
21     return 0;
22 }
```

Program 19: Write a program to find ASCII value of the character user inputs.

findASCIIValue.c

```

1  /* Program to find ASCII value */
2
3  #include <stdio.h>
4
5  int main() {
6      char c;
7
8      printf("Enter a character: ");
9
10     // Reads character input from the user
11     scanf("%c", &c);
12
13     // %d displays the integer value of a character
14     // %c displays the actual character
15     printf("ASCII value of %c = %d", c, c);
16     return 0;
17 }
```


Program 20: Write a program to output numbers in certain formats using printf() function.

formattedOutput.c

```

1  /* Prints int and float values in various formats */
2
3  #include <stdio.h>
4
5  int main(void)
6  {
7      int i;
8      float x;
9
10     i = 40;
11     x = 839.21f;
12
13     printf("|%d|%5d|%-5d|%5.3d|\n", i, i, i, i);
14     printf("|%10.3f|%10.3e|%-10g|\n", x, x, x);
15
16     return 0;
17 }

```

Program 21: Write a program to add two fractions taken from the user and show the result.

addTwoFraction;

```

1  /* Adds two fractions */
2
3  #include <stdio.h>
4
5  int main(void) {
6      int num1, denom1, num2, denom2, result_num, result_denom;
7
8      printf("Enter first fraction: ");
9      scanf("%d/%d", &num1, &denom1);
10
11     printf("Enter second fraction: ");
12     scanf("%d/%d", &num2, &denom2);
13
14     result_num = num1 * denom2 + num2 * denom1;
15     result_denom = denom1 * denom2;
16
17     printf("The sum is %d/%d\n", result_num, result_denom);
18
19     return 0;
20 }

```

Program 22: Write a program to add two fractions taken from the user and show the result.

upcCodeCheck.c

```

1  /* Computes a Universal Product Code check digit */
2
3  #include <stdio.h>
4
5  int main(void) {
6      int d, i1, i2, i3, i4, i5, j1, j2, j3, j4, j5;
7      int first_sum, second_sum, total;
8
9      printf("Enter the first (single) digit: ");
10     scanf("%d", &d);
11     printf("Enter first group of five digits: ");
12     scanf("%d%d%d%d%d", &i1, &i2, &i3, &i4, &i5);
13     printf("Enter second group of five digits: ");
14     scanf("%d%d%d%d%d", &j1, &j2, &j3, &j4, &j5);
15
16     first_sum = d + i2 + i4 + j1 + j3 + j5;
17     second_sum = i1 + i3 + i5 + j2 + j4;
18     total = 3 * first_sum + second_sum;
19
20     printf("Check digit: %d\n", 9 - ((total - 1) % 10));
21
22     return 0;
23 }
```

Program 23: Write a program to display the marksheet of the students.

studentMarksheet.c

```

1  /*Evaluate total, average and percentage of a student*/
2
3  #include<stdio.h>
4
5  int main() {
6      int a,b,c,d,e;
7      float total,average,percentage;
8
9      printf("Enter marks of 5 subjects : \n");
10     scanf("%d %d %d %d %d",&a,&b,&c,&d,&e);
11
12     total = a+b+c+d+e;
13     average = total/5;
14     percentage = (total/500)*100;
15
16     printf("Total marks = %f\n", total);
17     printf("Average marks = %f\n",average );
18     printf("Net percentage = %f\n",percentage);
19     return 0;
20 }
```


Program 24: Write a program to check the type of input character given by user.

checkCharacterInput.c

```

1 //program to check type of input character.
2
3 #include<stdio.h>
4
5 int main() {
6     char ch;
7
8     printf("Enter a character: ");
9     scanf("%c",&ch);
10
11     if((ch >= 65 && ch <= 90) || (ch >= 97 && ch <= 122)){
12         printf("Entered character is an alphabet");
13     }else if (ch >= 48 && ch <= 57) {
14         printf("Entered character is a digit");
15     }else {
16         printf("Entered character is a special symbol");
17     }
18     return 0;
19 }

```

Program 25: Write a program to check if the year given by user is a leap year or not.

isLeapYear.c

```

1 /*Program checks if the given year is a leap year or not*/
2 #include <stdio.h>
3
4 int main() {
5     int input,i;
6
7     printf("Please input a year and I will tell you if it's a leap year: ");
8     scanf("%d",&input);
9
10    if((input % 4 == 0 && input % 100 != 0) || input % 400 == 0) {
11        printf("Your input (%d) IS a leap year.\n\n",input);
12    } else {
13        printf("Your input (%d) is NOT a leap year.\n\n",input);
14    }
15
16    return 0;
17 }

```

Program 26: Write a program to check if the year given by user is a leap year or not using ternary operator.

checkCharacterInput.c

```

1  /*Program to check for leap year using ternary operator*/
2
3  #include<stdio.h>
4
5  int main() {
6      int y;
7
8      printf("Please input a year and I will tell you if it's a leap year: ");
9      scanf("%d",&y);
10
11     //Using ternary operator
12     (((y % 100 != 0) && (y % 4 == 0)) || (y % 400 == 0)) ?
13         printf("Your input (%d) is a leap year.",y) :
14         printf("Your input (%d) is NOT a leap year.",y);
15
16     return 0;
17 }

```

Program 27: Write a program to check the greatest number given by user.

greatestNumber.c

```

1  /*Greatest number using nested if statement*/
2
3  #include<stdio.h>
4
5  int main()
6  {
7      int a,b,c;
8      printf("Enter three integer number: ");
9      scanf("%d %d %d",&a,&b,&c);
10
11     if(a>b) {
12         if(a>c)
13             printf("%d is greatest", a);
14         else
15             printf("%d is greatest", c);
16     } else {
17         if(b>c)
18             printf("%d is greatest", b);
19         else
20             printf("%d is greatest", c);
21     }
22     return 0;
23 }

```